

Daniel Rajakumar

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Experience

Manufacturing Excellence Intern, Software Engineering, dsm-firmenich – Princeton, NJ Jun 2026 – Present

- Replaced 22 tracking spreadsheets with a Next.js and TypeScript application for 50+ employees across 7 business units and 8 manufacturing sites, centralizing assessment rubrics, element scoring, and reviewer submissions.
- Cut data-entry time by 2 hours per assessment cycle and reduced invalid records by 40% by designing an Excel import pipeline with ExcelJS that validates and migrates 500+ records per upload.
- Secured the app with company single sign-on and no separate user store by integrating Microsoft Entra ID through NextAuth v5, resolving an Edge-runtime failure by splitting provider config out of the auth middleware, and enforcing email-driven manager/reviewer role gating on every protected route.

Projects

RockyGPT – Campus AI Chatbot | rockygpt.danielrajakumar.com Jan 2026 – Present

- Built a full-stack campus assistant with Next.js 16, React 19, and TypeScript that answers questions on dining, events, directories, transit, housing, and safety, returning every answer with linked sources at a 2.1s median response time.
- Eliminated redundant embedding cost across releases by engineering a source-grounded retrieval pipeline on PostgreSQL/pgvector using 1,536-dimension embeddings with content-addressed chunk storage that re-embeds only when source text or model changes.
- Raised intent-classification accuracy from 72.6% to 83.9% and outcome accuracy to 100% by building an eval harness spanning 2,200+ scored queries and 43 automated test suites covering retrieval safety, privacy, and rate limiting.

VC370 Assembler and Emulator | github.com/daniel-rajakumar/VC370 Oct 2024 – Dec 2024

- Built a two-pass assembler and emulator in C++17 for the VC370, a 13-opcode accumulator architecture, translating symbolic assembly into 6-digit machine words executed across a 10,000-word memory space.
- Implemented a symbol table resolving forward references on the second pass, with four assembler directives (ORG, DC, DS, END) handling program origin, constants, and storage reservation.
- Structured the toolchain into 7 components with assemble-time and runtime error detection spanning illegal opcodes, undefined and multiply-defined labels, reserved-word collisions, memory-bound violations, and division by zero.

Skills

Languages: TypeScript, JavaScript, Python, C++, Java, SQL, HTML/CSS

Frameworks & Libraries: Next.js, React, Node.js, NextAuth, Tailwind CSS, Playwright, ExcelJS

Databases & Cloud: PostgreSQL, pgvector, Redis, AWS S3, Microsoft Entra ID

Tools & APIs: Git/GitHub, OpenAI API, REST APIs, embeddings/RAG

Education

Ramapo College of New Jersey – B.S. in Computer Science Jun 2026

Presidential Scholarship Recipient (full-tuition merit award)

Relevant Coursework: Computer Vision, Data Analysis & Visualization, Machine Learning, Web App Development

Leadership

Lead, Google Developer Student Club – Ramapo College Aug 2023 – May 2025

- Organized the college's first DevFest for 50+ students, coordinating a keynote and hands-on technical workshops.
- Directed a team of 8 to deliver 4+ coding events reaching 100+ total attendees, including a Java workshop.

Founding President, Computer Science Club – Ramapo College Apr 2023 – May 2024

- Founded the club and organized 10+ workshops on React, portfolio development, and collaborative software practices.
- Built and maintained the club website in HTML, CSS, and JavaScript, serving events and resources to 200+ students.